

Modelling, Dynamics and Kinematics

Final Exam

6 November 2024

13:45 – 16:10

(13:45 – 16:45 for extra time)

Formula sheet

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \Leftrightarrow \tilde{x} = \begin{pmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{pmatrix} \quad (1)$$

$$r = \frac{\omega \times v}{\|\omega\|^2}; \quad \lambda = \frac{\omega^\top v}{\|\omega\|^2} \quad (2)$$

$$\text{Ad}_{H_a^b} = \begin{pmatrix} R_a^b & 0 \\ \tilde{p}_a^b R_a^b & R_a^b \end{pmatrix}; \quad \text{ad}_{T_a^{c,b}} = \begin{pmatrix} \tilde{\omega}_a^{c,b} & 0 \\ \tilde{v}_a^{c,b} & \tilde{\omega}_a^{c,b} \end{pmatrix} \quad (3)$$

$$e^A := I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots \quad (4)$$

$$\theta \tilde{\omega} \in \mathfrak{so}(3), \quad \|\hat{\omega}\| = 1 \Rightarrow e^{\theta \tilde{\omega}} = I + \tilde{\omega} \sin \theta + \tilde{\omega}^2 (1 - \cos \theta) \quad (5)$$

$$e^{\tilde{T}} = \begin{pmatrix} e^{\tilde{\omega}} & \frac{1}{\|\omega\|^2} ((I - e^{\tilde{\omega}})(\omega \times v) + (\omega^\top v) \omega) \\ 0 & 1 \end{pmatrix} \quad (6)$$

$$\hat{T}_{\text{rot}} = \begin{pmatrix} \hat{\omega} \\ r \times \hat{\omega} \end{pmatrix}; \quad \hat{T}_{\text{tr}} = \begin{pmatrix} 0 \\ \hat{v} \end{pmatrix}; \quad \hat{T}_{\text{scr}} = \begin{pmatrix} \hat{\omega} \\ r \times \hat{\omega} + \lambda \hat{\omega} \end{pmatrix} \quad (7)$$

$$(\dot{\text{Ad}}_{H_b^a}) = \text{Ad}_{H_b^a} \text{ad}_{T_b^{b,a}} \quad (8)$$

$$\text{ad}_{T_{\bullet}^{b,\bullet}} = \text{Ad}_{H_a^b} \text{ad}_{T_{\bullet}^{a,\bullet}} \text{Ad}_{H_b^a} \quad (9)$$

$$(\dot{\mathcal{P}}^i)^\top = \text{ad}_{T_i^{i,0}}^\top (\mathcal{P}^i)^\top + (W^i)^\top \quad (10)$$

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau^\top \quad \left(G(q) = \frac{\partial V}{\partial q} \right) \quad (11)$$

Part I (Test)

Read carefully. There is no limit in the number of right or wrong answers in a question. Wrong answers will be penalised.

1. Tick true statements:

- ☐ Vector components are independent of the choice of basis.
- ☐ Having chosen a basis on a real vector space one can calculate angles between vectors.
- ☐ One must use an orthonormal basis to guarantee that the result of the application of a covector on a vector (e.g. $\alpha(v)$) is independent of the basis choice.
- ☐ For any linear map $\varphi: V \rightarrow W$, the dual map $\varphi^*: V^* \rightarrow W^*$ can be uniquely defined.
- ☒ If components of vectors $v \in V$ transform as $\tilde{v} = A \cdot v$ under a change of basis, then components of covectors $\alpha \in V^*$ transform as $\alpha = \tilde{\alpha} \cdot A$, where components of vectors are stacked into columns and components of covectors into rows.
- ☐ For any vector $v \in V$, there is a unique dual vector $v^* \in V^*$ such that $v^*(v) = 1$.

2. How well do you recognize tensors? Tick the true statements:

- ☒ $\text{Ad}_{H_i^j}$ is a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ -tensor defined on the Lie algebra $\mathfrak{se}(3)$.
- ☒ A wrench W is a $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ -tensor on $\mathfrak{se}(3)$.
- ☐ A homogeneous matrix $H_i^j \in \text{SE}(3)$ is a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ -tensor.
- ☒ A rotation matrix $R_i^j \in \text{SO}(3)$ is a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ -tensor on the vector space $\mathbb{R}^3$.
- ☒ A geometric Jacobian $J(q)$ is a linear map between two vector spaces.
- ☒ A mass matrix $M(q)$ is a $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ -tensor on $T_q\mathcal{Q}$ since $\frac{1}{2}\dot{q}^\top M(q) \dot{q}$ is the scalar kinetic energy of the system.

3. Tick the true statements:

- ☐ For a rigid motion h_2^1 in $\mathcal{E}$ mapping a frame Ψ_1 to a frame Ψ_2 , one has $\psi_1 \circ h_2^1 = \psi_2$. Here $\psi_i: \mathcal{E} \rightarrow \mathbb{R}^3$ (for $i \in \{1, 2\}$) is the map returning coordinates of points in space $\mathcal{E}$ in frame Ψ_i .
- ☐ Upper-left 3×3 block of H_a^b is a skew-symmetric matrix.
- ☐ Homogeneous matrix H_a^b represents the motion from frame Ψ_a to frame Ψ_b expressed in Ψ_b .
- ☒ A homogeneous matrix H_a^b is an element of a 6-dimensional manifold.
- ☒ The map $\psi_b \circ \psi_a^{-1}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a change of coordinates from a frame Ψ_a to a frame Ψ_b .
- ☐ $(H_a^b)^{-1} = (H_b^a)^\top$.

4. Consider the Lie group $SE(3)$. What is true about the space $T_H^*SE(3)$, that is the cotangent space of $SE(3)$ at $H \in SE(3)$?
- ☐ There is a duality pairing between a twist $T \in \mathfrak{se}(3)$ and an element of $T_H^*SE(3)$, and the resulting scalar represents mechanical power along a rigid body motion.
 - ☒ It is a 6-dimensional vector space.
 - ☐ It has a natural inner product.
 - ☒ There is a duality pairing between an element of $T_H^*SE(3)$ and the velocity $\dot{H}$ at H of any curve on $SE(3)$ passing through H .
 - ☒ It is the dual vector space of the space $T_HSE(3)$.
 - ☒ One can construct a map from $\mathfrak{se}^*(3)$ to $T_H^*SE(3)$ which will be a linear isomorphism.
5. Tick the true statements:
- ☒ $\dot{H}_a^b$ being right-translated to the identity is the twist expressed in Ψ_b .
 - ☐ Twist of body i moving relative to body j expressed in any coordinate frame fixed to the body j is the same.
 - ☒ Using the Brockett's product of exponentials formula, one can compute $H_i^0(q)$ of the i -th link in the middle of a serial-chain mechanism.
 - ☐ The unit twist $\hat{T}_i^{0,(i-1)}$ of the i -th joint is always constant.
 - ☒ The dimension of $\mathfrak{so}(2)$ is 1 and the dimension of $\mathfrak{se}(2)$ is 3.
 - ☐ $\tilde{T}_a^{c,b} = H_c^d \tilde{T}_a^{d,b} H_d^c$.
 - ☒ $T_a^{c,b} = -T_b^{c,a}$.
 - ☐ The chain rule for twists states that $T_n^{0,0} = T_1^{0,0} + T_2^{1,1} + \dots + T_n^{(n-1),(n-1)}$.
6. Consider a rigid body with a fixed point. The body's motion is described by $R(t)$ with $R(0) = I$. Tick the true statements.
- ☒ Torque (or moment) τ applied to the rigid body is an element of $T_I^*SO(3)$.
 - ☐ There always exists an element of $SO(3)$ that mirrors the rigid body.
 - ☒ $R^{-1}\dot{R}$ and $\dot{R}R^{-1}$ lie in the same vector space.
 - ☒ The expression $\exp(\dot{R}(0))$ is well-defined and returns an element of $SO(3)$.
 - ☒ Rodrigues' formula provides a closed-form expression for $\exp(\tilde{\omega}\theta)$ for any $\tilde{\omega} \in \mathfrak{so}(3)$ with $\|\omega\| = 1$ and $\theta \in \mathbb{R}$.
 - ☐ Denoting the angular velocity by $\tilde{\omega}(t) = \dot{R}(t)R^\top(t)$, one can always obtain the trajectory of the body as $R(t) = e^{\tilde{\omega}t}$.
 - ☒ At any instant of time, there are infinitely many points of the body with zero instantaneous velocity.

7. Tick the true statements:

- ☐ In the Brockett's formula, one can change the order of the exponentials since it is a product.
- ☒ $v_b^{r,r}$ in the twist $T_b^{r,r}$ is the relative velocity of the point of the body Ψ_b that is currently at the origin of the reference frame Ψ_r .
- ☒ Given a rigid body with a twist $T = (\omega, v) \in \mathfrak{se}(3)$, all the body-fixed points on the instantaneous screw axis have instantaneous linear velocity $\lambda \omega$.
- ☐ The geometric Jacobian acts on a wrench expressed in the end-effector frame.
- ☐ The geometric Jacobian is a linear isomorphism from $T_q \mathcal{Q}$ to $\mathfrak{se}(3)$.
- ☐ The geometric Jacobian is a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ -tensor.
- ☐ The geometric Jacobian has always a non-null determinant if the robot has six or more degrees of freedom.

8. Tick the true statements:

- ☒ The Euler-Lagrange equation of a robot is an equation on the cotangent space $T_q^* \mathcal{Q}$.
- ☒ Its right-hand side $\tau^\top$ comprises all external forces, such as motor torques and interaction forces.
- ☒ The mass matrix of a manipulator $M(q)$ is always an invertible square matrix.
- ☐ $M(q)$ is a $(0, 2)$ -tensor on the space of twists $\mathfrak{se}(3)$.
- ☒ The dynamics of a rigid body $(\dot{\mathcal{P}}^i)^\top = \text{ad}_{T_i^{i,0}}^\top(\mathcal{P}^i)^\top + (W^i)^\top$ is an equation on the space $\mathfrak{se}^*(3)$.
- ☐ This equation is valid only in an inertial frame, like Newton's law for a point mass $\dot{p} = F$.
- ☐ Consider a rigid body motion with respect to a fixed environment. There always exists a frame, called the principal inertia frame, which is fixed to the environment and where the matrix representing the inertia tensor of the rigid body is diagonal.

Part II (Open question)

Solving the differential kinematics of a serial chain mechanism with the geometric Jacobian involves computation of the unit twists of the joints expressed in the base frame Ψ_0 in a current configuration q . While simple mechanisms allow for these unit twists to be computed by inspection – that is, by examining and deducing the motion of the mechanism visually – the complexity of this task increases significantly as the number of joints and degrees of freedom grows. In such cases, an algorithmic approach is more suitable.

- (a) Detail, in a step-by-step manner, the algorithm you would develop to compute the configuration-dependent unit twists entering the geometric Jacobian $J(q)$ for a serial-chain mechanism with n simple joints. The input of the algorithm should consist of constant configuration-independent unit twists and homogeneous matrices in the reference configuration $q = 0$.

Begin by describing the procedure to obtain the inputs of the algorithm for a given mechanism starting from the assignment of the link-fixed frames. Then proceed with the steps to obtain the geometric Jacobian $J(q)$ as a function of the configuration q . Your description should include both textual and mathematical formulation of each step.

- (b) Compare and contrast this algorithmic method with the method of inspection.

Solution:

1. We count links from 0 to n and joints from 1 to n . Assign a frame Ψ_i to the i -th link placing it at the $i + 1$ -th joint with the *next*, $i + 1$ -th link, such that one of its axes coincides with the axis of the joint (for $i \in \{0, \dots, n - 1\}$). For rotational and screw joints this is the axis of the rotation, and for translational joints this is any line in the direction of translation.

Note that we are free in the choice of a frame Ψ_n for the last, n -th link. It may be used to represent the pose of the end-effector.

2. Specify the constant unit twists $\hat{T}_i^{(i-1), (i-1)}$ for each joint. This is a unit twist of the i -th link relative to the $(i - 1)$ -th link expressed in Ψ_{i-1} . Since this frame is chosen to be aligned with this joint, for each joint type the unit twist will have the following simple form:

Suppose, without loss of generality, that the axis of a rotational joint is aligned with the x -axis of a link frame, the axis of a screw joint (with pitch λ) is aligned with the y -axis, and the axis of a translational joint is aligned with the z -axis. Then the corresponding unit twists would be:

$$\hat{T}_{\text{rot}} = \begin{pmatrix} \pm 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \hat{T}_{\text{scr}} = \begin{pmatrix} 0 \\ \pm 1 \\ 0 \\ 0 \\ \pm \lambda \\ 0 \end{pmatrix} \quad \hat{T}_{\text{tr}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \pm 1 \end{pmatrix}$$

We simply put a ± 1 (and $\pm \lambda$ for screws) in the corresponding slot with the sign chosen based on the positive direction of the joint coordinate q^i (use the right-hand rule for screws).

3. Choose a convenient reference configuration $q = 0$.

4. Specify the homogeneous matrices $H_i^{i-1}(0)$ for all adjacent frames in the chosen reference configuration based on robot's parameters (for $i \in \{1, \dots, n\}$).

Note that the last homogeneous matrix $H_n^{n-1}(0)$ is not used in the algorithm for differential kinematics, and thus, may not be specified, although it would still be relevant for direct kinematics.

5. Using the specified unit twists and homogeneous matrices, compute the *configuration-dependent* homogeneous matrices $H_i^{i-1}(q)$ using the kinematic pair relation

$$H_i^{i-1}(q) = \exp(\tilde{T}_i^{(i-1), (i-1)} q^i) \cdot H_i^{i-1}(0).$$

6. Use the chain rule for homogeneous matrices to iteratively calculate the configuration-dependent homogeneous matrices $H_i^0(q)$ relating links to the inertial frame Ψ_0

$$H_i^0(q) = \prod_{k=1}^i H_k^{k-1}(q) = H_1^0(q) \cdot H_2^1(q) \cdot \dots \cdot H_i^{i-1}(q)$$

for each $i \in 1 \dots n - 1$.

7. Use the obtained configuration-dependent homogeneous matrices to transform all the unit twists $\hat{T}_i^{(i-1), (i-1)}$ to the inertial frame obtaining the unit twists in the *current configuration* $\hat{T}_i^{0, (i-1)}(q)$

$$\hat{T}_i^{0, (i-1)}(q) = \text{Ad}_{H_{i-1}^0(q)} \hat{T}_i^{(i-1), (i-1)}.$$

Alternatively, we may perform the same transformation in the matrix form as

$$\tilde{T}_i^{0, (i-1)}(q) = H_{i-1}^0(q) \cdot \tilde{T}_i^{(i-1), (i-1)} \cdot (H_{i-1}^0(q))^{-1}.$$

8. Finally, compose the twists into the Jacobian matrix $J(q)$ to solve the differential kinematics problem.

$$J(q) = \begin{pmatrix} \hat{T}_1^{0,0}(q) & \hat{T}_2^{0,1}(q) & \dots & \hat{T}_n^{0, (n-1)}(q) \end{pmatrix}$$

The inputs for the algorithm include **homogeneous matrices** $H_i^{i-1}(0)$ **in a reference configuration** which incorporate geometric parameters of the robot, and the constant **unit twists** $\hat{T}_i^{(i-1), (i-1)}$ **expressed in the link-fixed frames** which are determined by joint types. Providing such inputs requires *minimal effort* since they incorporate only *local* information about the robot (i.e. on adjacent links only) in a chosen *reference configuration*.

In contrast, the inspection method requires global information on robot's kinematics and necessitates visualising the robot's motion or resolving intricate trigonometric equations. Universality of the algorithm allows application to mechanisms with an arbitrary number of links and joint types, making it particularly suited for programming and automation.

However, while this method simplifies and standardises calculations for complex mechanisms, performing it manually may be difficult since it involves computing twist multiplications and matrix exponentials. The inspection method may still be preferable for simpler mechanisms where a quick, intuitive understanding of the motion is feasible.

Part III (Exercise)

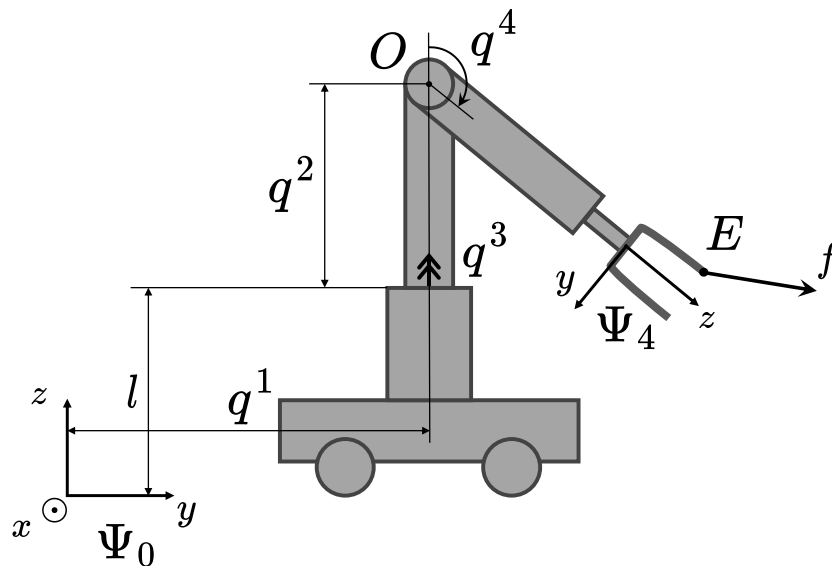
Consider the 4-DoF robot, shown in the figure. The x -axis of Ψ_0 points out of the page. The robot consists of:

- A translational joint (q^1);
- A translational joint (q^2);
- A rotational joint (q^3) with the axis parallel to the z -axis of Ψ_0 ;
- A rotational joint (q^4) with axis parallel to the x -axis of Ψ_0 at the configuration shown in the figure;

The frame Ψ_4 is attached to the end-effector. The coordinates of point O in Ψ_4 are $(0, 0, -a)$, and the coordinates of point E in Ψ_4 are $(0, -1, 2)$.

Indicate your choice for the reference configuration ($q^i = 0$, $i \in \{1, 2, 3, 4\}$).

1. Give Brockett's equation for direct kinematics and all unit twists that you need to calculate $H_4^0(q)$. *Note: There is no need to write out the matrix exponentials, but you must calculate the vector products that appear inside your expressions.*
2. Compute the geometric Jacobian $J(q)$ as a function of q and write an expression to find the twist of the end-effector. *Note: You must calculate the vector products that appear inside your expressions.*
3. Compute the velocity of the point E in the base frame Ψ_0 as function of q and $\dot{q}$.
4. Suppose an external force f is applied to the end-effector at the point E . Its components in Ψ_4 given by $f^4 = (b \ c \ d)$, i.e. the force is constant in the end-effector frame. This could be the case if e.g. a propeller mounted on the end-effector would produce a constant thrust. First, determine the induced wrench of the end-effector, and then compute the generalised forces τ that are experienced by the joints as the effect caused by this force as a function of q and $\dot{q}$.
5. How would you compute the induced wrench of the end-effector if the force was given in the base frame Ψ_0 ? E.g. in case the robot is pushing the wall.



Solution:

1. Let us consider that q^3 in the configuration shown in the figure is zero and choose the reference configuration accordingly.

Brockett's formula:

$$H_4^0(q) = e^{\tilde{T}_1^{0,0} q^1} e^{\tilde{T}_2^{0,1} q^2} e^{\tilde{T}_3^{0,2} q^3} e^{\tilde{T}_4^{0,3} q^4} H_4^0(0), \text{ where}$$

$$\begin{aligned} \hat{T}_1^{0,0}(0) &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} & \hat{T}_2^{0,1}(0) &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} & \hat{T}_3^{0,2}(0) &= \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} & \hat{T}_e^{0,3}(0) &= \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ -l \\ 0 \end{pmatrix} \\ H_4^0(0) &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l+a \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

2.

$$T_4^{0,0} = J(q) \dot{q}$$

$$J(q) = (\hat{T}_1^{0,0}(q) \quad \hat{T}_2^{0,1}(q) \quad \hat{T}_3^{0,2}(q) \quad \hat{T}_e^{0,3}(q)) \text{ , where}$$

$$\begin{aligned} \hat{T}_1^{0,0}(q) &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} & \hat{T}_2^{0,1}(q) &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} & \hat{T}_3^{0,2}(q) &= \begin{pmatrix} 0 \\ 0 \\ 1 \\ q^1 \\ 0 \\ 0 \end{pmatrix} & \hat{T}_4^{0,3}(q) &= \begin{pmatrix} \hat{\omega} \\ r \times \hat{\omega} \end{pmatrix} \end{aligned}$$

$$\hat{\omega} = \begin{pmatrix} -\cos q^3 \\ -\sin q^3 \\ 0 \end{pmatrix} \quad r = \begin{pmatrix} 0 \\ q^1 \\ l+q^2 \end{pmatrix} \quad \Rightarrow \quad \hat{T}_4^{0,3}(q) = \begin{pmatrix} -\cos q^3 \\ -\sin q^3 \\ 0 \\ (l+q^2) \sin q^3 \\ -(l+q^2) \cos q^3 \\ q^1 \cos q^3 \end{pmatrix}$$

3.

$$E^0 = H_4^0(q) E^4$$

$$\dot{E}^0 = \dot{H}_4^0(q) E^4 + H_4^0(q) \dot{E}^4 \stackrel{0}{=} \tilde{T}_4^{0,0} H_4^0(q) E^4 = \widetilde{(J(q)\dot{q})} H_4^0(q) E^4$$

4. The force f is applied at point E , in general, it induces a moment (torque) about the origin of the frame Ψ_4 . Thus, the wrench induced by the force f expressed in Ψ_4 consists of the force components f^4 and the moment components $r \times f^4$, where r is any vector from the origin of Ψ_4 to the line of action of the force f . For example, we can choose r to be the coordinates of the point E in Ψ_4 : $r = (0, -1, 2)$.

$$\begin{aligned} W^4 &= (\tau^4 \quad f^4) = \left((r \times (f^4)^\top)^\top \quad f^4 \right) = \left(\left(\begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} b \\ c \\ d \end{pmatrix} \right)^\top \quad f^4 \right) \\ &= (d - 2c \quad 2b \quad b \quad b \quad c \quad d) \end{aligned}$$

$$W^0 = W^4 \text{Ad}_{H_0^4}$$

$$\tau = W^0 J(q) = W^4 \text{Ad}_{H_0^4} J(q),$$

where $H_0^4(q) = (H_4^0(q))^{-1}$ is obtained from (1) and $J(q)$ is obtained from (2).

5. If the force is given in the base frame Ψ_0 as $f^0 = (b \quad c \quad d)$, then we first transform it to the end-effector frame Ψ_4 as $f^4 = f^0 R_4^0(q)$ and then proceed as in (4). The rotation matrix $R_4^0(q)$ can be obtained from $H_4^0(q)$.

Alternatively, we can directly compute the wrench induced by the force expressed in Ψ_0 . The moment of f about the origin of Ψ_0 is $r \times f^0$, where r is any vector from the origin of Ψ_0 to the line of action of the force f . For example, we can choose r as:

$$\begin{pmatrix} r \\ 1 \end{pmatrix} = E^0 = H_4^0(q) E^4.$$

Then the wrench of the end-effector expressed in Ψ_0 is

$$W^0 = \left((r \times (f^0)^\top)^\top \quad f^0 \right),$$

and the generalised forces τ in the joints are

$$\tau = W^0 J(q).$$